Simulation of Pectus Dynamics for Cardio-Respiratory Impact

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GitHub Repository/Shareable Resources:

UCI CHOC Cardiopulmonary GitHub Repository
CHOC x UCI MDS Public Google Drive

1 Introduction and Problem Statement

A common thoracic condition among adolescent and young adult patients is Pectus Excavatum (PE). PE is a chest wall deformity where the breastbone and ribs grow inward, creating a concave or sunken appearance, occurring in 1/300 births (Abdullah). Symptoms range from mild pain and exercise intolerance to cardio-thoracic impacts that increase cardiac risks. Surgical treatment may be required to correct the chest deformity, which involves the placement of 1-3 bars in the chest wall, akin to ribs, to reconfigure the chest outwardly. Typically, surgery is indicated if a common metric, the Haller Index (HI), derived from CT/MRI imaging, exceeds a certain threshold (HI > 3.25) (Knipe). The Haller Index is a ratio between the widest horizontal distance across the inside of the ribcage and the shortest distance between the sternum and the vertebra. A larger HI would indicate a more severe deformity.

Besides Haller Index, there are more physiologic parameters that are relevant to cardiopulmonary impairment. Parameters such as lung and heart compliances, airway resistances, cardiac chamber elastances, and pressure-volume relationships are important to understand the kinematics of the system, yet they cannot be measured directly without invasive procedures. The objective of this project, in collaboration with Children's Hospital of Orange County, CHOC, is to address this gap by developing a dynamic simulation of the cardiopulmonary system (heart and lung models) that integrates readily available patient data, such as demographics, MRI, echocardiography, and Cardiopulmonary Exercise Testing to obtain the aforementioned parameters. Furthermore, by combining simulation results with data analytics, we aim to uncover latent phenotypes within pectus excavatum patients. This allows a more nuanced understanding of heterogeneity and provides insights of compensatory mechanisms by which the patients adapt their cardiopulmonary dynamics.

2 Related Work

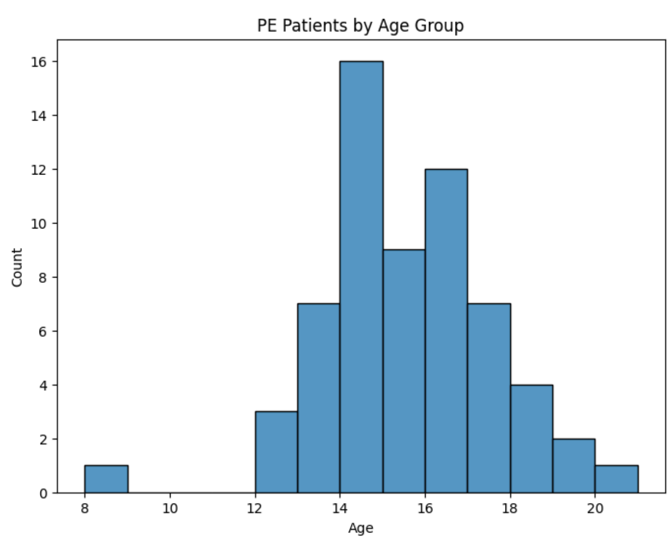
Simulating cardiopulmonary dynamics through modeling has been an active research for several decades by implementing computational frameworks to simulate respiratory and cardiac mechanics (Niederer et al., 2019). For example, Sarmiento and Guerrero (2020) built a Respiratory Mechanics Simulink GUI which provides kinematics model of lung capable of estimating pressures, flows, resistances, and compliances throughout cardio-respiratory systems by interactive simulation in MATLAB/Simulink. Through a model called Cardiovascular System in Simscape™ with ECMO Machine, Nord (2023) offered a representation of heart dynamics, simulating cardiac chambers, valves, and segment sizes using Simscape. Despite multiple computational frameworks in the cardiopulmonary system, specific studies in analyzing pectus excavatum conditions remain limited. Nicola, et al. (2021) examined the physiological assessment of pectus excavatum patients while exploring postoperative simulation using vacuum bell procedure. While procedures to alleviate pectus conditions remain outside of this project scope, this journal, along with aforementioned publicly available simulation models from Sarmiento and Guerrero (2020) and Nord (2023), serves as methodological basis for our analysis of cardiopulmonary function in pectus excavatum patients.

3 Datasets

The dataset consists of clinical and physiological information of 62 patients diagnosed with Pectus Excavatum. It is made up of demographic information as well as measurements from a variety of clinical tests including Cardiac Magnetic Resonance Imagery (Cardiac MRI), Echocardiograms, and results from Cardiopulmonary Exercise Tests (CPET).

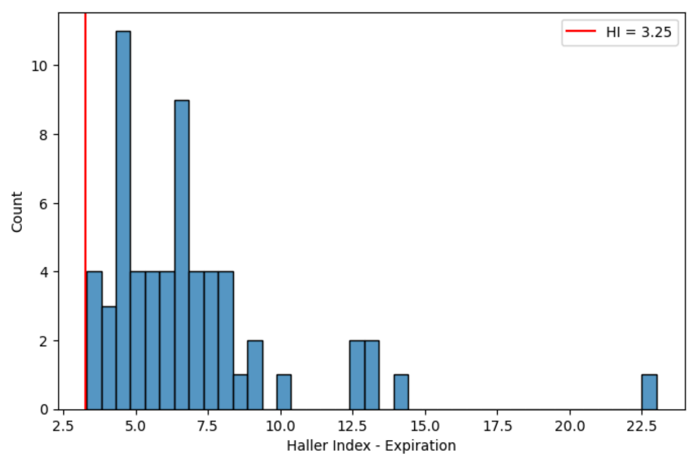
Typical demographic information was provided, including age, weight, height, gender, race, ethnicity, and body surface area (BSA). Most of the patients in our dataset are in their teens as typically surgical intervention is not required until adolescence. We were also provided information on whether the patient had surgical intervention as well as indicators on whether heart and lung symptoms appeared and if they were subsequently resolved after surgery.

From the Cardiac MRI, we were provided with measurements extracted from the scans of the



heart. This includes standard metrics used to support the diagnosis of PE, including the Haller Index and the Cardiac Compression Index (CCI), another measurement looking at the dimensions of the heart and chest to help diagnose PE. All patients in our dataset are above the Haller Index threshold of 3.25, which typically indicates the need for surgical intervention. Also included are important measurements on the function and size of the heart. This includes the end diastolic volume index (EDVi) and end systolic volume index (ESVi) of both ventricles in the heart. This measures the volume of these chambers of the heart at the end of the heart's contraction (systole) and relaxation phase (diastole) standardized based on the patient's BSA. Other measurements include the ejection fraction percentage (EF%) and cardiac index (CI) of each ventricle. The ejection fractions measures the percent of blood pumped out of the chamber with each heartbeat, while the Cardiac Index measures the amount of blood pumped each minute normalized for BSA.

The Echocardiogram, an ultrasound of the heart, provided additional metrics of the heart's performance. This included the EF % measured through the ultrasound, the Left Ventricular Internal Diastolic Dimension (LVIDd), a measure of the size of the left ventricle, and the E/e' ratio, a measure of diastolic function based on the velocity of blood flow and the speed at which the muscles in the left ventricle relax.



The Cardiopulmonary Exercise Tests (CPET) contained data on the performance of the patient during exercise. Measures include the max heart rate of the patient, as well as the patient's O2 Pulse and VO2 Max, which measure the ability of the body to consume oxygen. Additionally, the CPET results included important measurements about the functionality of the lungs. This includes the Forced Vital Capacity (FVC), the amount of air forcefully expelled after a deep breath, Peak Expiratory Flow (PEF), the maximum velocity of exhalation, as well as FEV1 and FEF 25-75. FEV1 is the Forced Expiratory Volume over 1 second, which measures the amount of air expelled over 1 second of a forceful exhalation. Similarly, FEF 25-75 measures the forced expiratory flow, or average airflow rate, over the middle half of a forced exhalation (25-75% of the exhalation period).

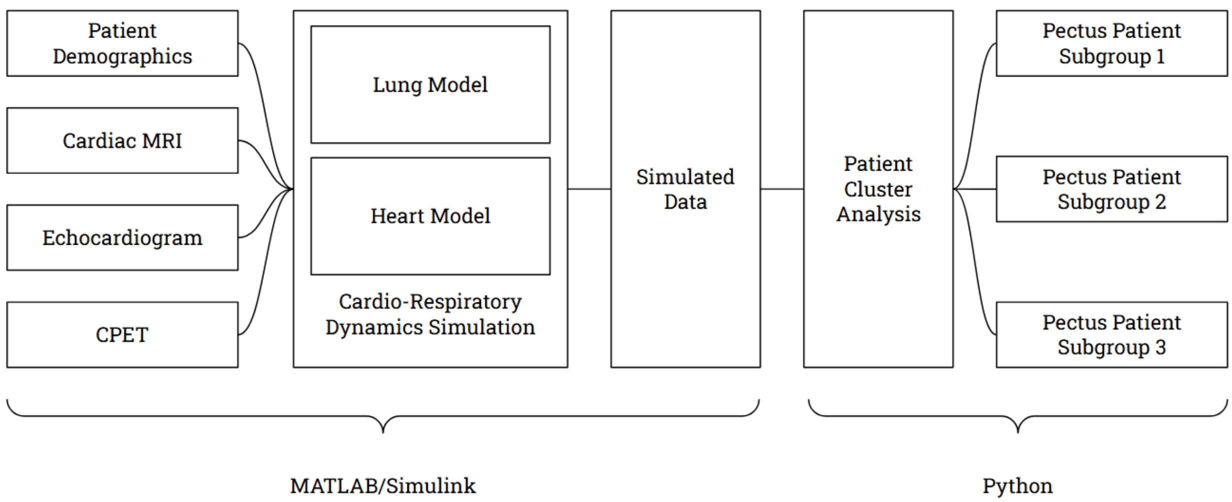
4 Technical Approach and Methods

Simulation Framework

To simulate heart and lung performance for each pectus patient, we can leverage publicly available MATLAB heart and lung simulation models to approximate individual cardiorespiratory behavior and organize patients into meaningful groups based on their simulated functional patterns and clinical measurements. The selected models rely on parameters such as system compliances and resistances, which aren't frequently recorded and often require invasive procedures to measure. To address this, we map model parameters using available information from provided clinical measurements and incorporate expert guidance from a pediatric cardiologist to ensure physiologically consistent simulations. Additionally, we used language models, such as ChatGPT and Claude, to assist in constructing MATLAB scripts, helping streamline model implementation and data processing.

Because our study is exploratory and limited in time, we prioritize understanding the dynamics of simulated heart and lung metrics rather than achieving high simulation accuracy. Future iterations of the model will refine and improve these simulations to better reflect typical pectus patient performance.

It is also essential to establish a baseline dataset of healthy patients for interpreting simulated outcomes. We generated a synthetic healthy dataset matching the size of the patient dataset using established population norms and adding Gaussian noise to introduce patient variability. This provides a reference that helps differentiate patterns specific to pectus from those expected in a typical healthy population. The overall pipeline from data ingestion to cluster analysis is visualized below.



Given the interdependence of heart and lung function, we model each system separately to capture distinct physiological mechanisms while maintaining a consistent parameter mapping framework across patients. Each model follows the same workflow: clinical features are mapped to model parameters, simulations are generated for each patient, and resulting metrics are used in a cluster analysis performed in Python to identify functional subgroups within our dataset. After both models have been evaluated, the simulated heart and lung metrics are combined into a unified dataset for a final clustering step, determining subgrouping based on joint cardiorespiratory behavior.

Lung Simulation

To simulate lung function, we use a model from Antonio Nariño University (Sarmiento and Guerrero, 2020) that represents lung mechanics under a mechanical ventilator, which supports or controls breathing by regulating airflow, pressure, and volume. The model's key parameters include lung, thoracic, and airway tissue compliance, which indicate how easily the lungs and chest wall expand and contract, and central and peripheral airway resistance, which reflect the degree of airflow obstruction in large and small airways. Additional parameters such as breathing frequency, PEEP, and peak inspiratory pressure can also be adjusted. Using a ventilator-based simulation ensures that all patients are assessed under consistent respiratory conditions, allowing fair comparisons of lung function and the effects of pectus deformities independent of individual breathing effort. In consultation with a pediatric cardiologist, we mapped patient dataset variables to model parameters to generate patient-specific inputs. We model the relationships between patient measurements and simulation parameters using a modified logistic (sigmoid) function:

$$S(x; c, w) = \frac{1}{(1+exp(-(x-c)/w))}$$

The sigmoid function captures non-linear physiological responses while keeping outputs within realistic bounds, preventing extreme or implausible values. It generates a multiplier coefficient that adjusts a baseline reference value for a healthy individual according to the patient's measured variable. Each parameter has its own set of sigmoid shape parameters (c,w, L) to control correlation, scaling, and limits. The lung model includes an additional parameter, γ, which controls how extreme patient measurements influence the lung compliance, which provides more variance across all patients. For example, the baseline value for lung compliance is computed using the height (Helliesen et al.), with the overall formula for a patient's lung compliance being:

$$LC = LC_0 \left(1 - L_{LC} * max(0, \frac{S(1; c_{FVC}, w_{FVC}) - S(FVC_{pct}; c_{FVC}, w_{FVC})}{S(1; c_{FVC}, w_{FVC})_{ref}})^{\gamma_{LC}} \right)$$

Thoracic compliance and airway resistances follow similar sigmoid-based equations, with their own coefficients. Default ventilator settings include a breathing frequency of 15 breaths per minute, PEEP of 5 cmH_2O, peak inspiratory pressure of 10 cmH_2O, and an inspiratory-to-expiratory ratio of 2. Full details regarding model parameter mapping, ventilator-setting definitions (such as tidal volume, respiratory rate, PEEP, and inspiratory pressure), and equation coefficients are provided in Appendix A.1–4.

Heart Simulation

To simulate cardiac function, we use the CardioVascularSystem Simscape model (Nord). The project first loads default healthy parameters, and for each PE patient, we overwrite key parameters using clinical measurements from CHOC. In consultation with a pediatric cardiologist, we designed a mapping from MRI, echocardiography, and CPET variables to model parameters such as ventricular volumes, arterial and venous compliances, and valve/loop resistances. This mapping is summarized in our heart parameter flowchart (Appendix B.1).

We first personalize chamber geometry and timing. Indexed volumes are converted to absolute volumes using body surface area and a similar approach is done for the right ventricle. Operating heart rate is reconstructed from CPET as approximately $HR \approx 0.6 \times HR_{max}$, or from percent-predicted max, $HR_{max}^{pred} \approx 220 - age$. This HR then sets the cardiac period and systolic duration,

$$t_c = \frac{60}{HR}, \quad t_s = 0.16 + 0.3\, t_c$$

so that each patient's simulation runs at an appropriate beat frequency. Measured systolic and diastolic blood pressures (SBP, DBP) and LV stroke volume are combined to estimate large-artery compliance,

$$PP = max(5, SBP - DBP), \ SV_{LV} = max(1, LV\,EDV - LV\,ESV), \ C_{art} \approx \frac{SV_{LV}}{PP}$$

which is used to rescale proximal and distal systemic compliances so simulated aortic pressures are consistent with brachial BP.

Pectus-specific effects enter through the Haller Index and compression flags. Higher HI reduces thoracic and venous/pulmonary compliances using a simple scaling factor

$$m_{HI} = max(0.5, 1 - 0.06\,(HI - 2.5)),$$

and inferior vena cava (IVC) and right-ventricular (RV) compression indicators increase corresponding venous and pulmonary resistances while slightly decreasing local compliances. Additional metrics such as LVEF, RVEF, Cardiac Compression Index, sternal torsion angle, septal E/e′ ratio, LV mass index, VO_2 max %, O_2 pulse %, and left atrial size are used as smooth multipliers (bounded between about 0.7 and 3.0) on valve resistances (aortic, pulmonary, mitral) and myocardial viscous terms on the LV and RV sides, allowing the model to reflect how impaired reserve and chest wall deformity alter forward flow, afterload, and filling pressures.

For each patient, we simulate multiple heartbeats and analyze the final beat to avoid transients. The model logs time series for aortic and left ventricular pressure, right atrial/venous and pulmonary pressures, and left/right inflow and outflow (LVOF, RVOF, LVIF, RVIF). Over the last cardiac period, we summarize each signal into scalar metrics (mean, maximum, minimum, standard deviation) and compute stroke volume and cardiac output as $SV = \int_{last\ beat} LVOF(t)\,dt, \ CO = \frac{SV \times HR}{1000}$ [L/min].

The same pipeline is applied to a synthetic healthy cohort generated from population norms, providing a baseline for comparison. These scalar heart metrics form the heart feature set used in our PCA and clustering analyses alongside lung and clinical variables. Full details regarding model parameter mapping are provided in Appendix B.2–5.

Clustering Discussion

To better understand latent patient subgroups, we aim to build an interpretable clustering model for individuals with PE using both simulated and observed clinical data. Our goal is to determine whether the patient population naturally separates into distinct subtypes or instead lies along a continuous severity spectrum.

Data Preparation and Principal Component Analysis

To obtain an interpretable representation of our high-dimensional dataset, we applied Principal Component Analysis (PCA) for dimensionality reduction before clustering. PCA helps us identify major axes of variation in the data, and by examining feature loadings, we can interpret each Principal Component (PC) and generate meaningful visualizations for downstream analysis.

Before performing PCA, we standardized all numerical features using the median and median absolute deviation to account for non-normal and skewed distributions. Categorical variables were properly encoded, and the dataset was checked for inconsistencies or missing values to ensure clean input for the analysis.

PCA was applied to the preprocessed dataset, reducing the original feature space to three principal components that capture the majority of the variance. To interpret these components, we computed the correlation matrix between the original features and the PCs. Identifying features with strong correlations to each component, we were able to characterize the underlying patterns that each PC represents (Penn State University, 2025).

The next step in the analysis was to determine the optimal number of patient subgroups (k). We evaluated clustering performance using the Elbow Method, Silhouette Score, and Calinski–Harabasz Index. While k = 2 produced the highest Calinski–Harabasz score, the elbow and silhouette scores suggested 3 as the ideal k value. Hence, we decided to go forward with k = 3.

Analyzing Clustering Methods to Categorize Subgroups

After establishing the optimal number of subgroups for analysis, we compared two widely used clustering methods: agglomerative and k-means, to identify which method most effectively captured patient symptom severity.

As an initial evaluation, we combined data from both PE patients and non-PE (healthy) patients to assess each method's ability to separate these distinct groups clearly. Upon completion, both clustering algorithms were able to distinguish between PE and healthy individuals, rendering this initial test insufficient for selecting an ideal clustering method.

To obtain a more discriminative evaluation, we next applied both clustering methods exclusively to the PE patient data and evaluated the results based on statistical characteristics across relevant physiological measures. This analysis provided a clearer distinction between the methods. K-means separated patients into subgroups with clear degrees of severity. In contrast, agglomerative clustering provided subgroups with less distinct interpretability. Based on these findings, k-means was deemed a more effective method for drawing meaningful conclusions from.

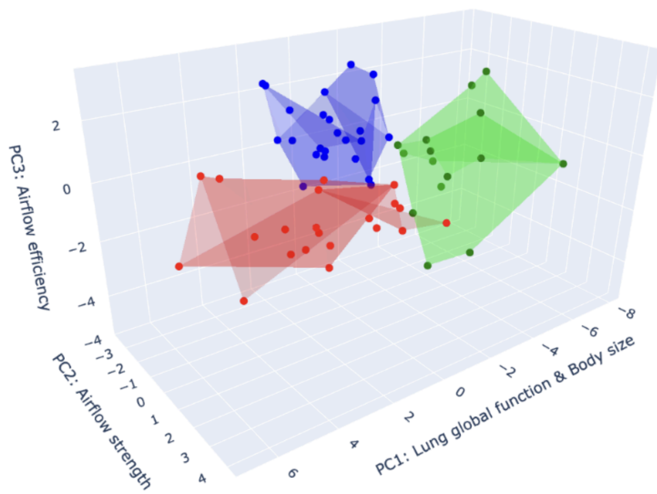
5 Evaluation/Validation/Experiments

To evaluate the accuracy and clinical relevance of our models, we validate simulation outputs against healthy patient simulations and then perform cluster analysis to explore latent patterns in the patient dataset. The primary focus of this analysis is to capture patient-specific behavior and provide insights into pectus deformity severity. Model validation and tuning were performed under a pediatric cardiologist's guidance to ensure that simulations reflect reasonably physiological responses.

Lung Simulations and Clustering

For the lung model, we generated simulations over time for lung volume, transpulmonary and alveolar pressures, respiratory flows, and dead space ventilation under standardized ventilator conditions. To compare across patients and allow clustering analysis, these signals are summarized into scalar metrics, such as mean, maximum, and standard deviation, representing overall magnitude, variability, and extremes of respiratory function. Baseline comparisons against our generated healthy dataset highlight characteristic differences in patients with pectus excavatum: alveolar flow is approximately 25% lower, indicating reduced gas exchange, while lung volume is around 30% higher, showing an increased effort in breathing to compensate for decreased air intake (see Appendix A.5 for plots comparing pectus and healthy patients).

To reduce dimensions for ideal clustering and plotting, we applied PCA to our simulation results and dataset. The first principal component (PC) reflects overall lung size and global function, the second PC reflects small airway performance and expiratory flow, and lastly, the third PC reflects airflow efficiency based on dead-space versus effective ventilation. More details on the principal components for the lung model can be found in Appendix C.1.



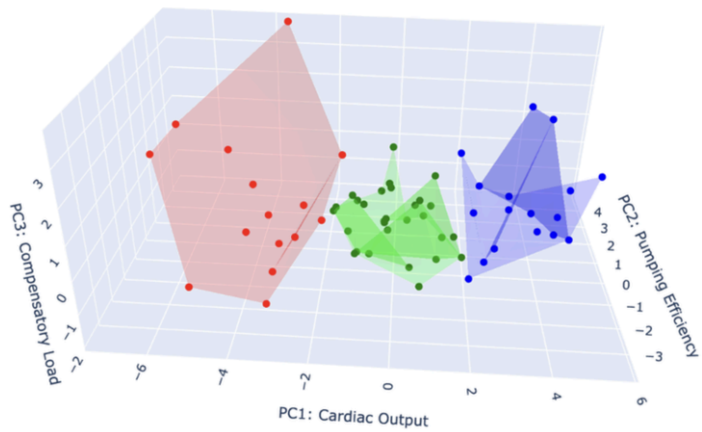
We then clustered the resulting reduced data to characterize our patients into descriptive subgroups. The first cluster (red) was characterized by strong lung size and global function, combined with high airflow strength and efficiency. Patients in this group generally exhibited the healthiest overall respiratory performance, resulting in milder PE severity. In contrast, the second cluster (green) was characterized by a reduced lung size and global function, but still had decent airflow strength and average airflow efficiency, although not as strong as the first cluster. These characteristics suggest that some lung and bodily structures were compromised within this group, but that airflow-related functions within this group's respiratory system were still able to be maintained. Patients in this group appear to represent a more average severity among our set of PE patients. Lastly, the third cluster (blue) exhibited a more average lung size and global function than the second cluster, but airflow metrics were substantially lower than the previous two groups. This group represented the poorest functional performance among the three clusters and represented the highest severity among PE patients in our measured data.

Taken together, the findings of our lung clusterings indicated that lung and bodily structure may have some indication on PE respiratory function, but that other non-lung factors may be involved in measuring the full severity of PE symptoms.

Heart Simulations and Clustering

For the heart model, we were able to simulate the pressure, volume, and amount of blood flow through the chambers of the heart. With different initial conditions, it took some time for the outputs of the simulation to reach a steady state. To address this we ran our simulations for 7 heartbeats for each patient, using the last heartbeat for feature generation. As with the lung model, we summarized the simulation results using scalar metrics, such as mean, maximum, and standard deviation, for each of the simulation outputs. Compared to our generated healthy dataset we notice characteristic differences in patients with PE, for example higher aortic pressure in PE patients (see Appendix B.6).

We then applied PCA to the metrics extracted from the simulations for our 62 patients and used the first 3 components for visualization and clustering. The first PC is related to the pressure and volume of the chambers of the heart, representing cardiac output, the amount of blood flowing through the heart. The second PC dealt with overall outflow from the ventricles and heart rate, while the third PC focused on the volume and load specifically for the right ventricle, which sits closer to the chest wall and may be more impacted than the left ventricle. These PCs explained over 80% of the variance of the data from our simulations. More details on the principal components for the lung model can be found in Appendix C.2.

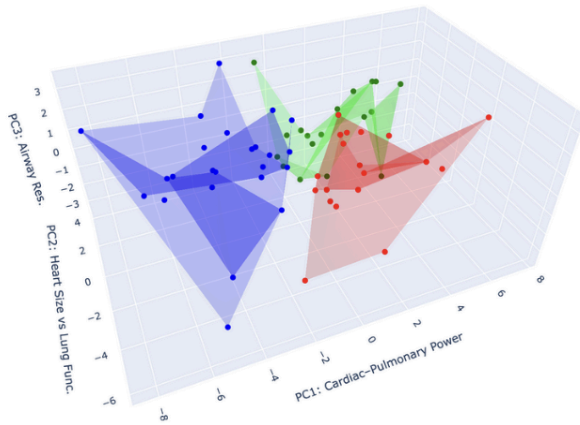


From the heart simulations, we found 3 main clusters of PE patients. The first cluster (red) had the worst performance in terms of heart function. The patients in this cluster were characterized by having lower heart rate as well as lower ventricular volumes. These patients had very low cardiac output as the low ventricle volumes mean less blood flowing per heartbeat, and a lower heart rate means less blood being pumped per minute. The patients in the second cluster (green) made up our intermediate severity group for the heart. While not as severe as the red cluster, these patients also had decreased heart function, but not to the same extent as those in the first cluster. These patients tended to have a more moderate heart rate and cardiac output. Meanwhile, the third cluster (blue) was the highest performing group. Within our dataset of PE patients, these patients had the strongest heart performance characterized by higher heart rates, ventricular volume, and stroke volume. With these higher heart rates and volumes, this means that the heart is both pumping more often but also ejecting more blood per heartbeat, which means more blood circulating through the body. So, despite the chest wall deformity, the heart was much less affected in this cluster compared to the others.

In addition to the lung, we find that the cardiovascular system may also be heavily affected by PE. While not all patients exhibited poor heart performance, our findings indicate that some PE patients may have lower overall cardiac output.

Combining Lung and Heart Simulations for Final Analysis

The final step in this project is to integrate lung and heart simulation outputs to identify phenotypes using a more holistic approach. Similar to previous clusterings, we applied PCA to our dataset of 62 patients with the addition of both lung and heart simulation results. The first PC is attributed to overall cardiopulmonary performance, driven by patients' cardiac output, aortic pressure, and lung volume. The second PC reflects the tradeoff between heart and lung performance, while the third PC captures small airway dynamics and the effectiveness of lung output. More details on the principal components for the combined model can be found in Appendix C.3.



These PCs collectively generate three clusters. Patients in the first cluster (red) indicate a more dominant lung performance. These patients have higher lung capacity but lower cardiac output. Patients in the second cluster (green) have the lowest cardiopulmonary performance of all patients with low lung capacity and low cardiac output. These patients have the most severe case of PE, with their cardiopulmonary performance restricted. The third cluster (blue) reflects a more dominant heart performance. These patients have lower lung capacity but higher cardiac output, exhibiting opposite patterns from patients in the red cluster.

Comparing these results to previous clustering based on heart or lung simulations individually, the combined model implies compensatory interactions between heart and lung to maintain cardiopulmonary performance despite physical impairment due to PE.

6 Team Member Participation

Member	**Key Responsibilities**
Alan Pham	- Researched correlations between measured metrics in dataset and unknown metrics unobtainable by sponsor - Implemented collection of ipynbs that provided analysis groundwork for project - 50% - Took leading role with class deliverables, contributing to the development of presentation slides and documents
Andrew Nguyen	- Researched prior medical literature and proposed key physiological variables to design a lung simulation framework, including a sigmoid-based parameter mapping method - Implemented MATLAB scripts to generate patient-specific lung simulations - Developed a synthetic healthy patient sampling method using population norms - Led planning, coordination, and organization to maintain steady project progress
Danny Suradja	- Development of heart model in Simulink and MATLAB script - Researched on parameter mapping from available dataset to feed into the model

	- Assisted with analysis and visualization of heart model simulation output and the PCA interpretation as well as clustering analysis
Hyun Woo (Harry) Choi	- Implemented the heart simulation pipeline in MATLAB/Simulink, including scripts to run patient specific and healthy cohort simulations using the Heart model - Assisted with analysis and visualization of heart simulation results and integrated heart-derived features into the joint heart–lung PCA and clustering pipeline
Minh Cao	- Cleaned, validated, and summarized clinical and simulated datasets to support downstream cardiopulmonary modeling - Implemented collection of ipynbs that provided analysis groundwork for project - 50% -Research interpretation of PCA results for cardiopulmonary features in PE patients, helping identify and characterize latent subgroups across heart and lung performance models
Tim Lanthier	- Conducted EDA and helped with visualizations on the initial dataset - Researched medical literature in order to estimate heart model parameters - Assisted with MATLAB scripts for the heart simulation - Adjusted heart model scripts in order to simulate heart function for healthy patients - Assisted with interpretation for heart and combined heart-lung PCA and clustering

Our team divided responsibilities across simulation and analysis domains: Andrew developed and executed the lung simulation pipeline, while Harry, Danny, and Tim focused on designing and implementing the heart simulation workflow. Minh and Alan conducted the clustering analyses for lung, heart, and combined simulation outputs and the CHOC-provided dataset. Together, these contributions formed a fluid workflow allowing us to cohesively perform our analyses.

7 Discussion and Conclusion

This study shows that dynamic simulations of heart and lung function can help identify meaningful subgroups of patients with PE. By combining each patient's clinical measurements with heart and lung simulations, we were able to find patterns that correspond to different levels of pectus severity, influenced by factors like body size and different compensatory responses. These results demonstrate how simulations can reveal hidden differences among patients with rare conditions, providing insights that are hard to obtain through standard clinical tests.

The project also represents a broad exploration into biomedical research, showing how simulations can uncover patterns in patient populations and illustrate the effects of structural deformities. There were many challenges and limitations faced along the way. Our team had limited background in human anatomy and little experience with MATLAB and Simulink, which required close guidance from physicians to ensure the models were realistic and the results meaningful.

Looking ahead, the models will be refined and expanded by researchers collaborating with CHOC. Planned improvements include combining the lung and heart models, creating a 3D chest wall model to show the pectus deformity, and developing 3D visualizations that translate simulation results into clear explanations for patients and families. These tools could help clinicians and families better understand how pectus affects breathing and heart function and support more personalized care. Overall, this project demonstrates the potential of simulations to reveal insights into rare pediatric conditions and lays the groundwork for future research.

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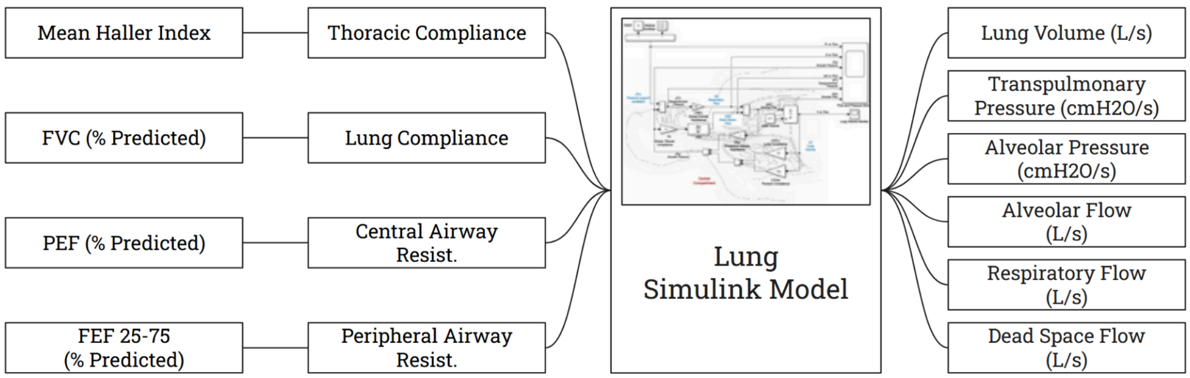
Appendix

Appendix A

1. Lung Model Simulation Diagram



2. Lung Model Parameter Mapping

Target Parameter	Dataset Variable	Dataset Variable Meaning	Number Format	Correlation
Lung Compliance	FVC (% Predicted)	Forced capacity in exhale	Percentage	Positive (+)
Thoracic Compliance	Haller Index (HI)	Chest wall deformity index	Numerical	Positive (+)
Central Airway Resistance	PEF (% Predicted)	Peak flow during exhale	Percentage	Negative (-)
Peripheral Airway Resistance	FEF25-75 (% Predicted)	Peak flow in the middle of exhale	Percentage	Negative (-)
Airway Tissue Compliance	N/A	N/A	N/A	Constant

3. Parameter Mapping Shape Parameters

Parameter	Coefficient c	Width w	Notes
LC (FVC%)	$c_{FVC} = 1.0$	$w_{FVC} = 0.10$	Controls sigmoid for lung compliance reduction
TC (HI)	$c_{TC} = 10.0$	$w_{TC} = 3.2$	Controls sigmoid for thoracic compliance scaling
CR (PEF%)	$c_{CR} = 1.0$	$w_{CR} = 0.10$	Controls sigmoid for central airway resistance scaling

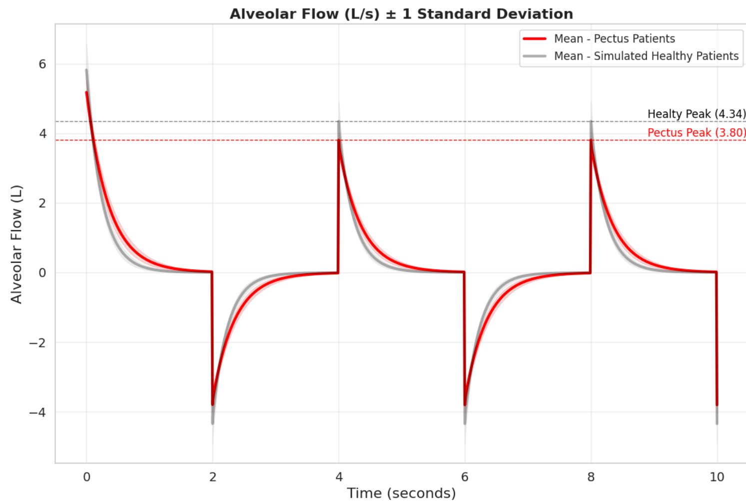
PR (FEF25-75%)	$c_{PR} = 1.0$	$w_{PR} = 0.10$	Controls sigmoid for peripheral airway resistance scaling

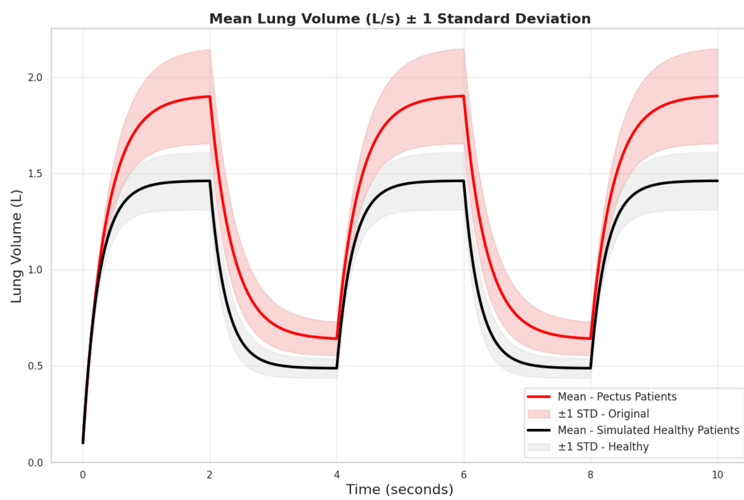
Parameter	Maximum adjustment	Notes
LC	$L_{LC} = 0.20$	Max 20% decrease
TC	$L_{TC} = 0.8$	Max 80% decrease
CR	$L_{CR} = 0.20$	Max 20% increase
PR	$L_{PR} = 0.40$	Max 40% increase
LC steepness	$\gamma_{LC} = 1.5$	Shape control for lung compliance curve

4. Default Ventilator Settings

Setting	Symbol	Value	Description
Breathing Frequency	FR	15 (breaths/min)	Ventilator setting for respiratory rate
Positive End-Expiratory Pressure	PEEP	5 (cmH2O)	Ventilator baseline pressure at the end of exhale
Peak Pressure	PP	10 (cmH2O)	Ventilator peak inspiratory pressure
Inhale:Exhale Ratio	E	2	Ratio of inspiratory time to expiratory time

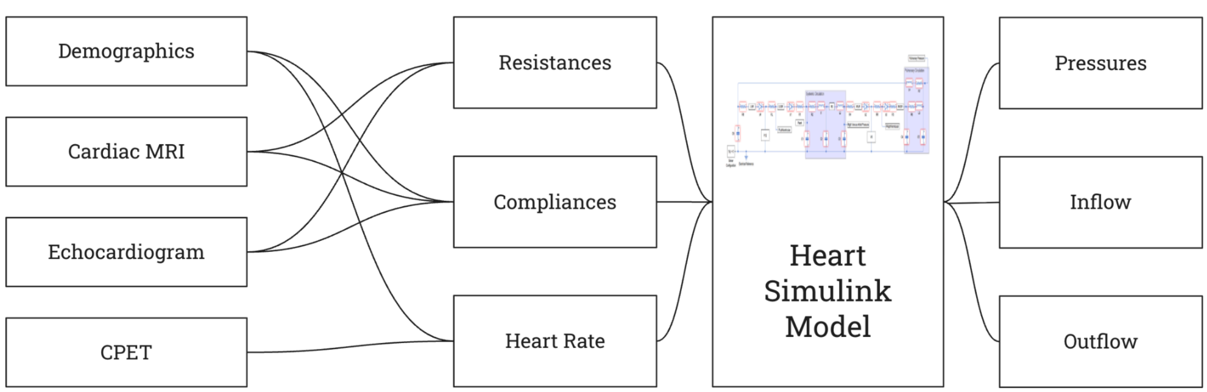
5. Lung Simulation Pectus vs. Healthy





Appendix B

1. Heart Model Simulation Diagram



2. Global Geometry, Timing, and Large-Artery Compliance

Target Model Group	Model Parameters	Main Dataset Variables	Approximate Relationship / Effect	Direction
LV / RV initial volumes	X120 (LV EDV), X60 (RV EDV)	LVEDVi, LVESVi, RVEDVi, BSA	LVEDV ≈ LVEDVi × BSA, LVESV ≈ LVESVi × BSA, RVEDV ≈ RVEDVi × BSA; set initial LV/RV volumes in model.	↑ LVEDVi or BSA → ↑ LVEDV/RVEDV
Stroke volume estimate	implicit (used for Cart)	LVEDVi, LVESVi, BSA	SV_LV ≈ max(1, LVEDV − LVESV).	↑ LVESVi → ↓ SV_LV (for fixed LVEDVi)
Heart rate & timing	HeartRate, tc, ts	HR max, HR max %predicted, age (or direct HR)	If HRmax available: HR ≈ 0.6 × HRmax. If only %pred: HR ≈ 0.6 × (HRmax%/100) × (220 − age). Then tc = 60/HR, ts ≈ 0.16 + 0.3·tc.	↑ HRmax or HRmax% → ↑ HR, ↓ tc
Systemic arterial compliance	C1, C2	SBP, DBP, LVEDVi, LVESVi, BSA	PP = max(5, SBP − DBP); SV_LV from above; Cart ≈ SV_LV / PP. Scale C1, C2 so C1 + C2 ≈ Cart (preserve C1:C2 ratio).	↑ SV_LV or ↓ PP → ↑ C1+C2

3. Thoracic / Venous / Pulmonary Compliances and Loop Resistances

Target Model Group	Model Parameters	Main Dataset Variables	Approximate Relationship / Effect	Direction
Thoracic / venous / pulmonary compliance	C3, C4, C5, C6	Mean Haller Index (HI)	Compute mult = max(0.5, 1 − 0.06·(HI − 2.5)). Then: $C3 \approx C3_0 \cdot (0.85 \cdot mult + 0.15)$, $C4 \approx C4_0 \cdot mult$, $C5 \approx C5_0 \cdot mult$, $C6 \approx C6_0 \cdot (0.9 \cdot mult + 0.1)$.	Higher HI → lower C3–C6 (down to ~50–60% of healthy)
Systemic venous return	R3, C3	IVC compression flag	If IVC compression present: $R3 \approx 1.2 \cdot R3_0$, C3 ≈ 0.9·C3 (harder venous return, less venous compliance).	IVC comp: ↑ R3, ↓ C3

Pulmonary arterial load	R6, C5	RV compression flag	If RV compression present: $R6 \approx 1.2 \cdot R6_0$, $C5 \approx 0.9 \cdot C5_0$ (higher pulmonary resistance, stiffer pulmonary arteries).	RV comp: ↑ R6, ↓ C5

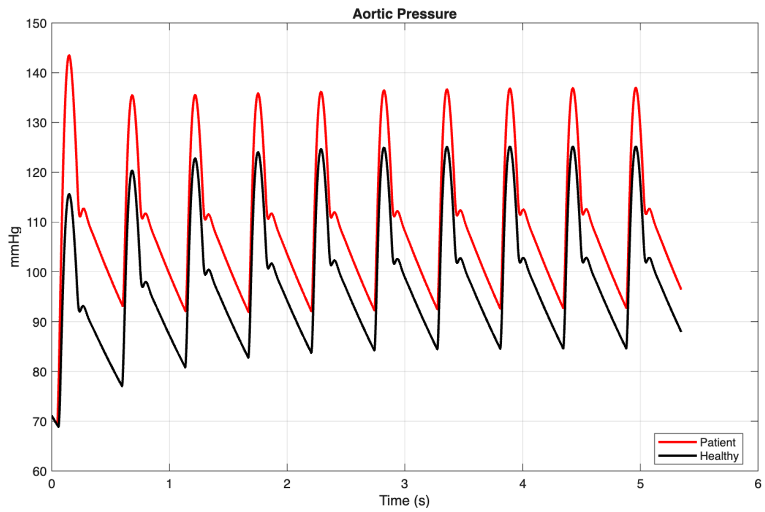
4. Valve Resistances

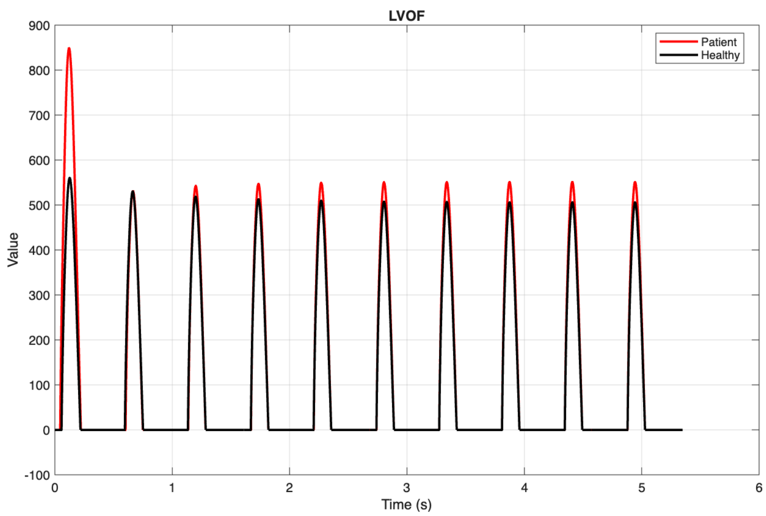
Target Model Group	Model Parameters	Main Dataset Variables	Approximate Relationship / Effect	Direction
Aortic valve resistance	R1	LVESVi, LVEF	Scale R1 by factor $\approx 1 + 0.3 \cdot [0.6 \cdot f(\text{LVESVi}) + 0.4 \cdot g(\text{LVEF})]$, where f,g are linearly increasing in high LVESVi / low LVEF and clipped.	↑ LVESVi, ↓ LVEF → ↑ R1 (up to ~1.3×)
Pulmonary / RV outflow valves	R4, R5	CCI, Sternal torsion angle, RV compression flag	Multiply R4,R5 by a composite factor (clipped ~0.7–2.0) that increases with higher CCI, larger torsion, and RV compression.	↑ CCI, ↑ torsion, RV comp → ↑ R4,R5
Mitral valve resistance	R8	E/e′ (septal)	R8 scaled by $\approx 1 + 0.2 \cdot h(E/e')$, where h increases with higher E/e′ (clipped).	↑ E/e′ → ↑ R8 (higher filling resistance)
Pulmonary venous side	R7	E/e′ (septal), LA size	R7 scaled by factor (clipped ~0.7–2.5) that increases with higher E/e′ and larger LA.	↑ E/e′ or ↑ LA size → ↑ R7

5. Myocardial “Stiffness” / Viscous Terms

Target Model Group	Model Parameters	Main Dataset Variables	Approximate Relationship / Effect	Direction
LV viscous term / stiffness	RL	E/e′ (septal), LV mass index, O_2 pulse %pred	RL scaled by product of factors (clipped ~0.7–3.0): increases with higher E/e′, higher LV mass index, lower O_2 pulse %.	↑ E/e′, ↑ LV mass, ↓ O_2 pulse → ↑ RL
RV viscous term / stiffness	RR	RVEF, O_2 pulse %pred	RR scaled by factors (clipped ~0.7–3.0): increases with lower RVEF and lower O_2 pulse %.	↓ RVEF, ↓ O_2 pulse → ↑ RR

6. Heart Simulation Pectus vs. Healthy





Appendix C

1. Lung PCA

PC #	Representation	Top Contributing Columns
1	Overall lung function + size vs PE severity	- Lung Function (FEV1) (+++) - Lung Compliance (LC), Lung Volume (++) - Haller Index (HI) (- -)
2	Performance of small-airway in the lung	- FEF25-75pct_post (+++) - Peripheral Resistance (- -) - Sim_AlvP_PeakMin (- -) *Measures how strong and how fast air move out of the*

		system during exhalation
3	How efficient is your airflow and how much is wasted (dead-space, inefficiency)	- Sim_DeadFlow_PeakMax (++) - Sim_Flow_PeakMax (- -), etc… *Measures how much air goes into "dead space", how efficient your air flow is*

2. Heart PCA

PC #	Representation	Top Contributing Columns
1	Pressure and Volume of heart chambers	- LV & Aortic Pressures (+++) - LV + RV Volume (End Diastolic & Systolic) (++) - LV Outflow (- -)
2	Heart rate and blood outflow from ventricles	- LV Outflow (+++) - Heart Rate - Haller Index (--)
3	Right Ventricle volume *RV typically more affected by PE due to its position closer to the chest*	- RV Volume (Diastolic & Systolic) (+) - Heart Rate (-)

3. Combined Heart and Lung PCA

PC #	Representation	Top Contributing Columns
1	Overall cardiopulmonary (heart & lungs) performance. *Captures the shared strength of heart pressure and flow dynamics together with lung volume and global respiratory performance*	- Left ventricular pressure variation(+++) - Left ventricular outflow (+++) - Aortic pressure (+++)
2	Heart and lung functional tradeoff. *Highlights an inverse pattern between heart chamber volumes and lung function.*	- Ventricular end-diastolic and end-systolic volumes (+++) - Lung function measures (− −) - Lung volume variation (− −)
3	Airway resistance and compensatory respiratory response. (How easy for a patient to breathe) *Separates patients based on airflow resistance, wasted ventilation, and the pressure required to move air through the system*	- Compression ratio (CR) (− − -) -Peak expiratory flow (PEFpct) (+++) - Alveolar pressure extremes (Sim_AlvP_PeakMin, Sim_AlvP_PeakMax) (+++)